LLm evaluation.

AI Engineer

An ai engineer is someone who builds applications and products on top of foundation models.

V.V.Imp

{ LLm evals. } { edge mindset }

answer

1) LLm evals why
2) RoadMap

The why

Do you create LLm Based Applications?

yes ✓

How do you evaluate it?

↳ Vibe testing.

↓

Means casually ~~typing~~ trying an LLm app with a few prompts & judging it by feel.

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~~Goal for this week~~

~~I'M eVals~~

Casestudy ①

Air canada AI Fiasco

chatbots

1) Bereavement Fare?

Book now refund later.

refund?

Customer care

No refund must request Before Travel

Judge

your Chatbot your responsibility

c$ 650.88 + Fees

Casestudy 2

chevy chatbot Fiasco:

dealer

emotionally convience

car $1

viral embarrassment.

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case study ③

chatgpt hallucinate → generate cases that does not cases:

lawyer represent fake case studies:

evaluation is super important

then why don't people test their llm applications.

* Because it is not straightforward.

Software testing V.S llm evals:

1) Deterministic: v.s probablistic:

For given input → op.

But llm app are Built on top of llm

llm → probablistic.

same input → different output

2) Correctness vs Multidimensional check

(2+2) = 4.

answer → Factual
→ Citation
→ Latency

aspects that may vary

llm evaluation is super important

Curriculum:

1) llm evals.
2) .llm evals landscape
3) llm evals → Benchmarks.
4) llm application evals.
5) eval pipeline.
6) Rag evals
7) Agent evals.
8) safety evals.
9) operation evals.

1 week goal

1) Master llm evaluation → campusX. 19 videos.

2) llm gateways & middleware - Krish Naik - 43 min

3) Loop & learness engineering - Krish Naik - 35 hr.

☆ 4) Active defence & Nemo guardrails. - 8 hours

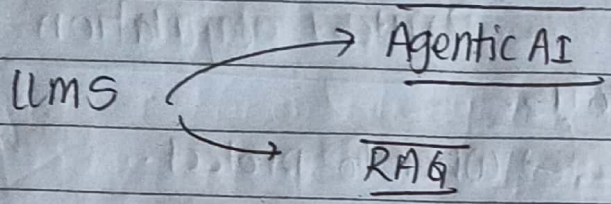
Nvidia Nemo guardrails.

5) the Capstone → FDE - 8 hr

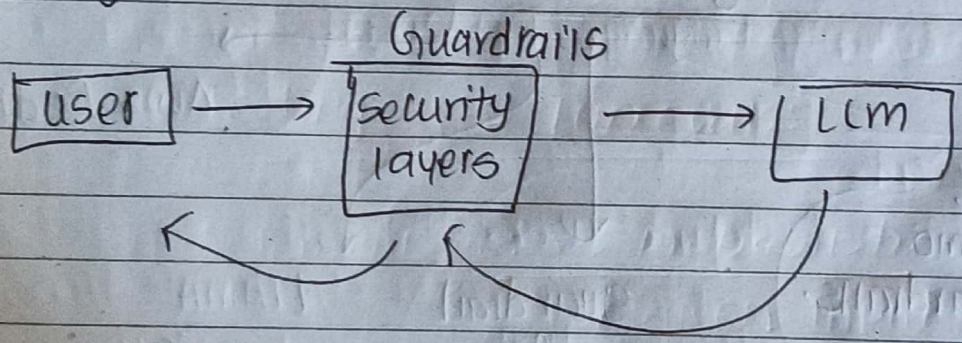
llm guardrails

↳ llm security ✓

Prompt injection , Jail break , Cost saving



~~Security~~



Robust, Fault tolerance → gateways.

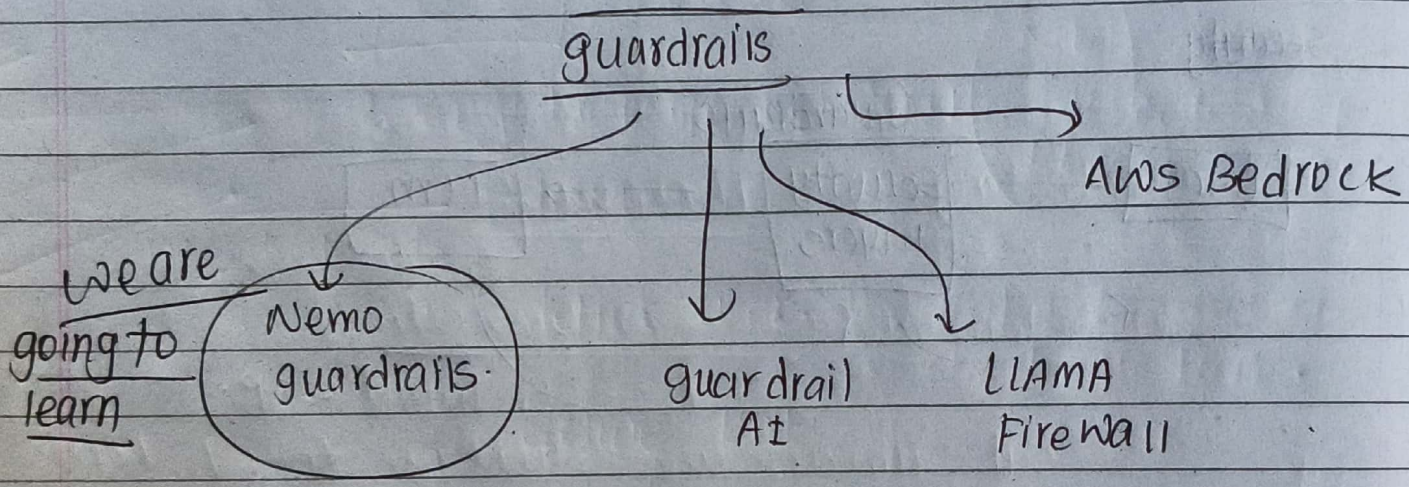
Security

↳ guardrails.

LLm gateways → litellm

Core capabilities

secured guardrails: → rules + regulation
↳ when to protect.



demo — prompt injection + Jail Break

when people go OFF topic

More token consumption

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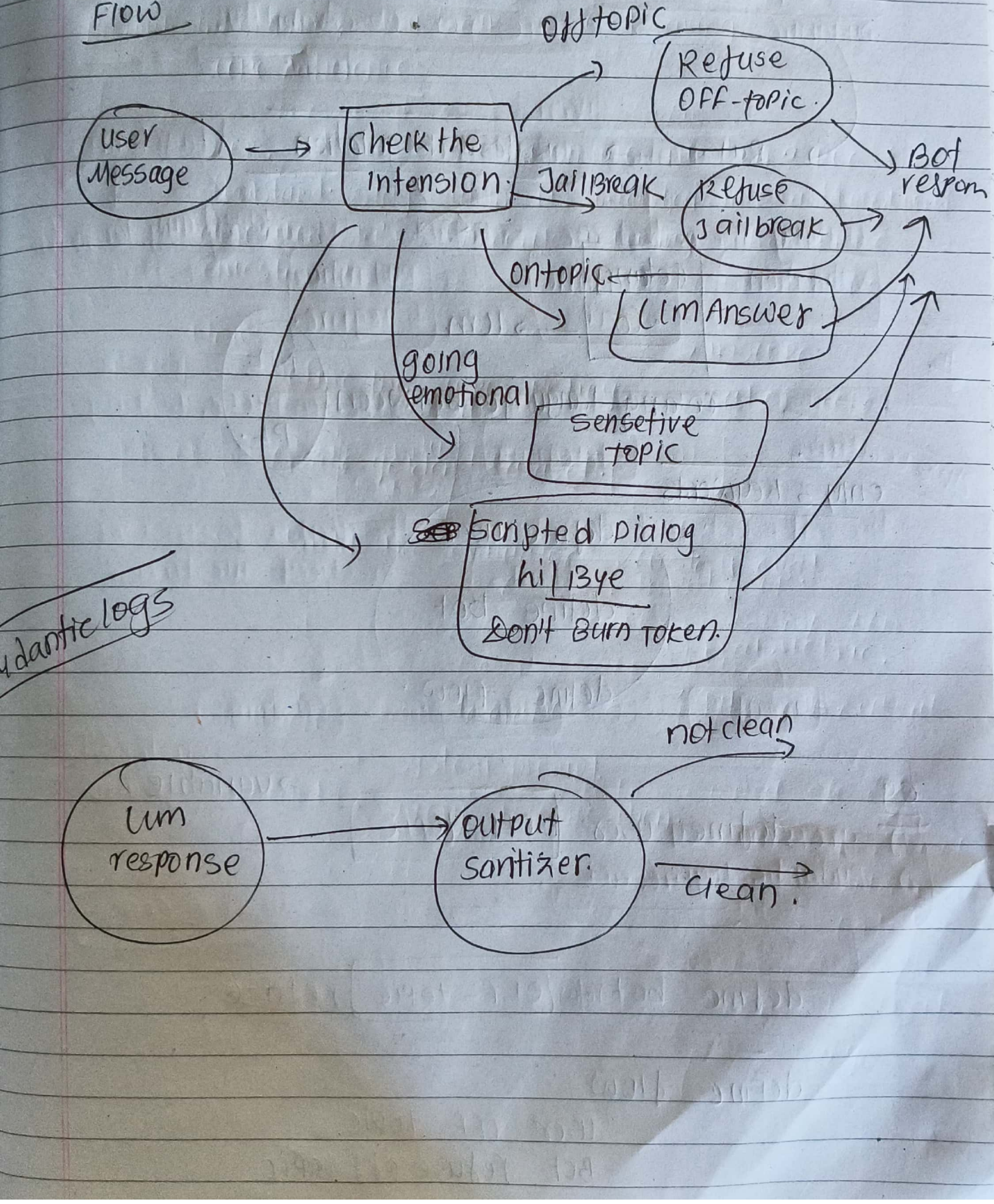
Message Flow



Nemo Guardrails

Rails → Rules & Regulation

Colang
- .Co → LLM

Colang language between NL & PL.

only 3 keywords

- define user
- define bot
- define flow

eg

define user OFF-topic → variable.

define bot ~~refuse~~ OFF-topic ~~refuse~~.

define flow
user OFF topic
Bot refuse off topic

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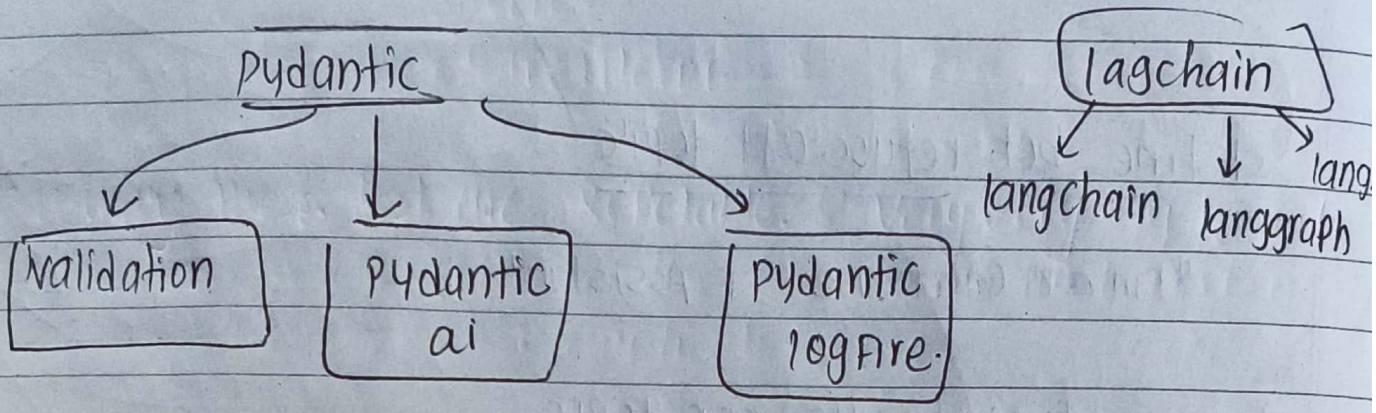
Forward deployed engineer

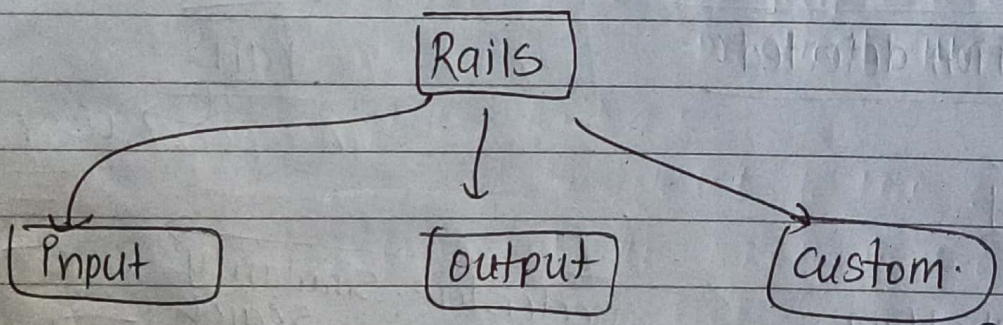
Nemoguardrails. → security layer
pydantic → logFire For observability:



#

AWS Bedrock

~~Telegram~~



systematic rails.

1) grok api key
2) pydantic logFire api

Llm evaluation nitish campusx:

what are llm evals?

are
1) systematic
2) Repetable.
3) clear criteria.

Llm evals:

Llm evals are systematic repetable tests used to judge an Llm or Llm-powered system against a clean criteria.

systematic: → datasets with every edge case.

Repetable: change prompts, model, retriever, chunking stratery.
We should be able to run the same test again & compare results.

clear criteria: Correct
Simple explanation.
grounded.

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LLm evals → Test → LLm | LLm Based Application

↓

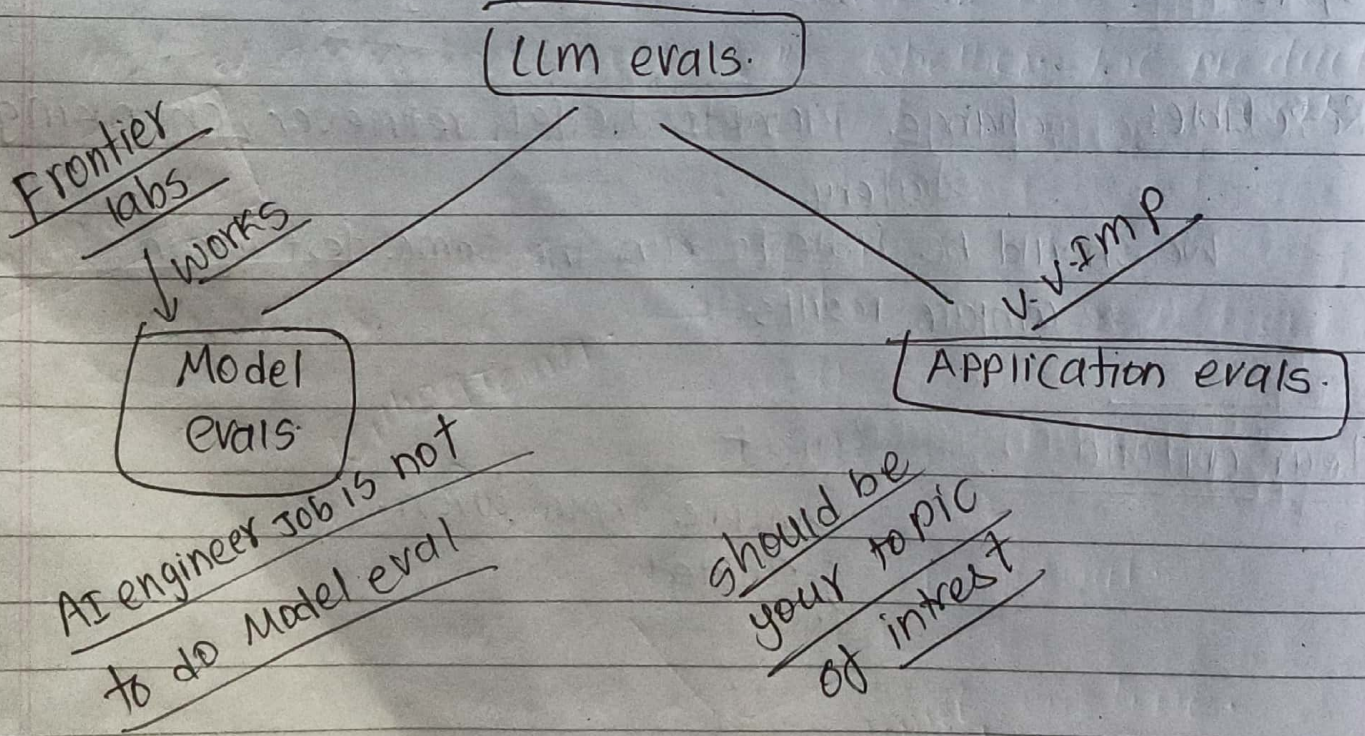
Test.

given
clear criteria.

& usually these test are Repetable & very very systematic.

An eval is not a Metric. An eval is the complete testing setup.

the goal is not just to get a score. the goal is to answer practical questions.



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Model evals:

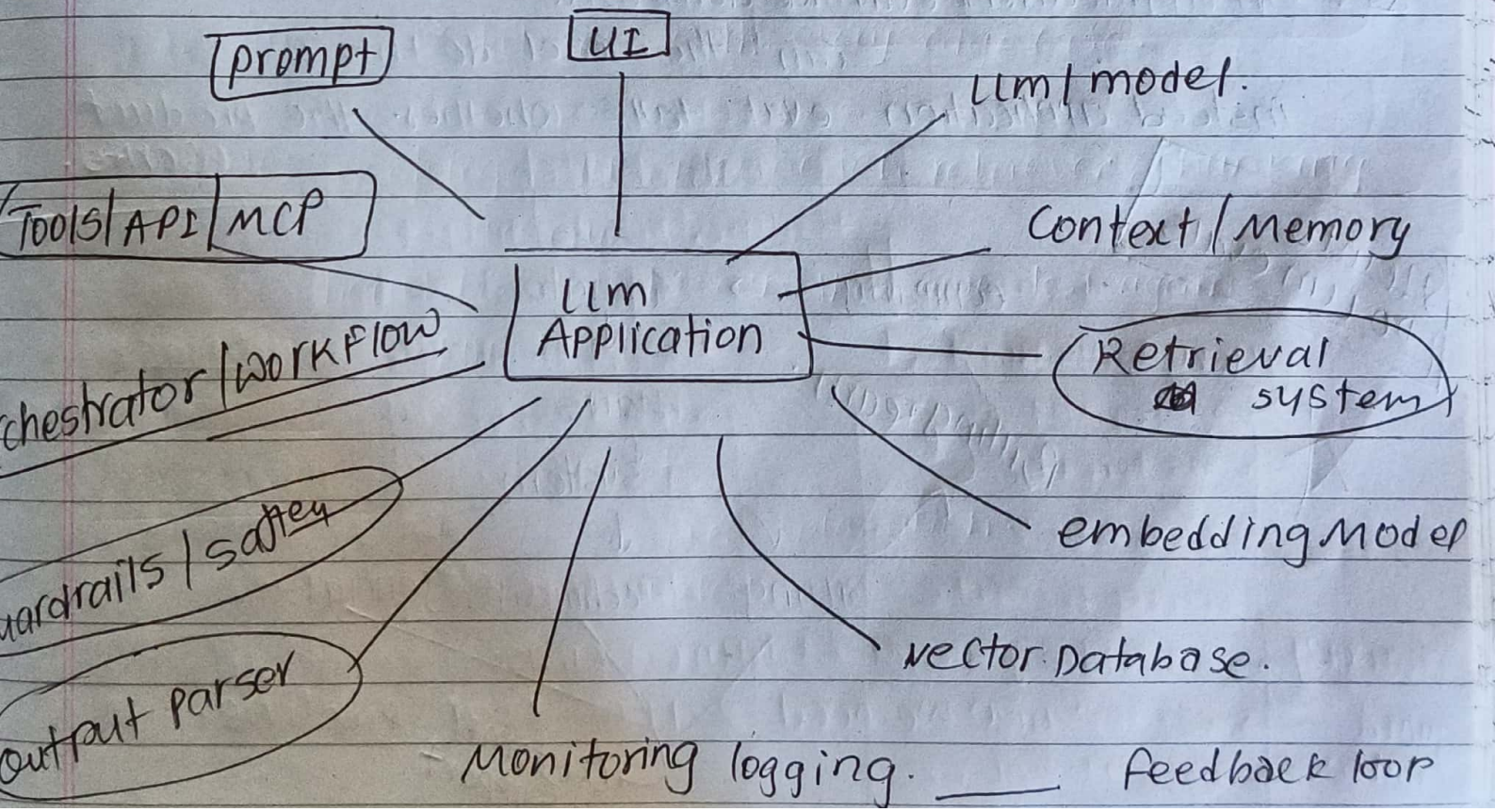
↳ evaluate the model itself.

↳ capabilities of a model

---

some important capabilities?

Capability Area	Famous Benchmark
Reasoning Knowledge	→ MMLU
Maths	— GSM8K
Coding	— HumanEval, SWE-Bench.
instruction Following	— IFEval
Long context	— Needle in a Heystack.
Multimodal Understanding	
Tool-use.	



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Analogy

Frontier labs

processor chip — smartphone

LLM — Application.

- Application evals access the behaviour and Performan of an LLM-powered application,

✓ wheather at the level of entire system.

or

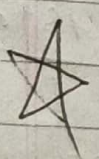
✓ specific component within it.

* in application eval we don't ask:

Can the model do this?

instead application eval tells whether the product works.

How to DO LLM evals



Company → Zomato

Billing Team.

Mails System.

Billing — Technical — General.

Customer Support

A single LLM Application can have Multiple evals.

#

email → LLM →

Before deployment we have to evaluate.

Flow

1) Define Task & Target.

↓

Define a success criteria.

↓

Prepare a Dataset. → Based upon Actual Data + Manual labelling → golden Dataset.

***

Make golden dataset Rich

Add Data to the golden Dataset.

↓

Define an eval method.

- Can be Automated python code
- Human
- LLM.

↓

Run the model ← golden Dataset again.

↓

evaluate the results

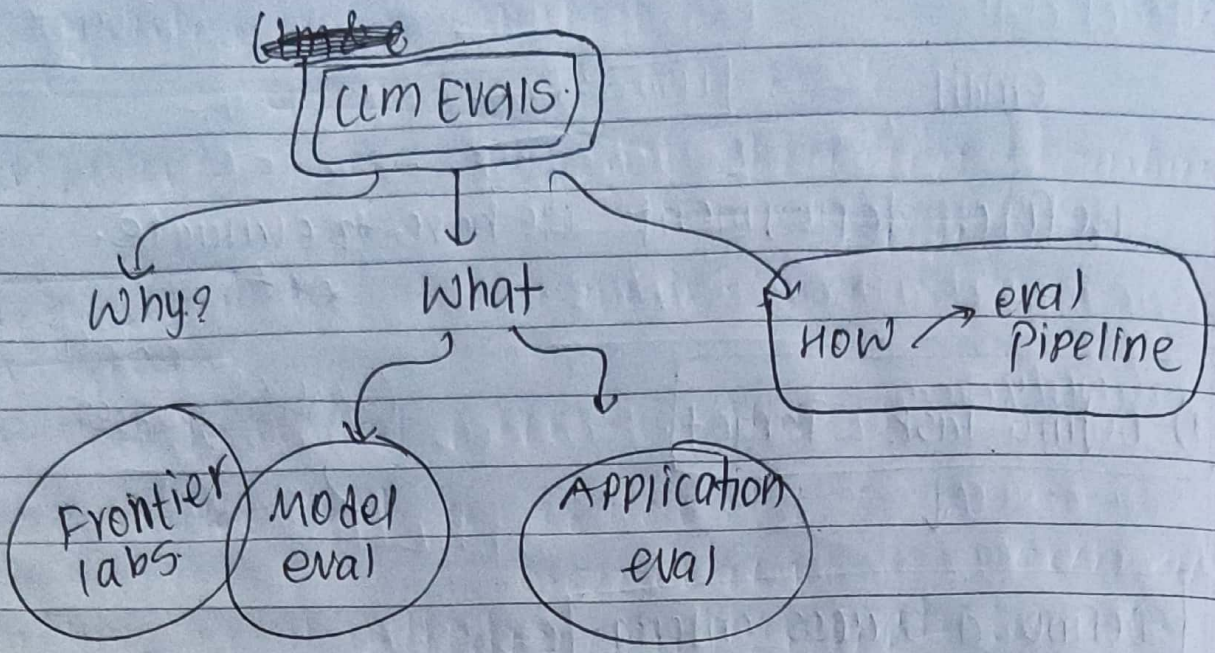
↓

Analyze the results.

↓

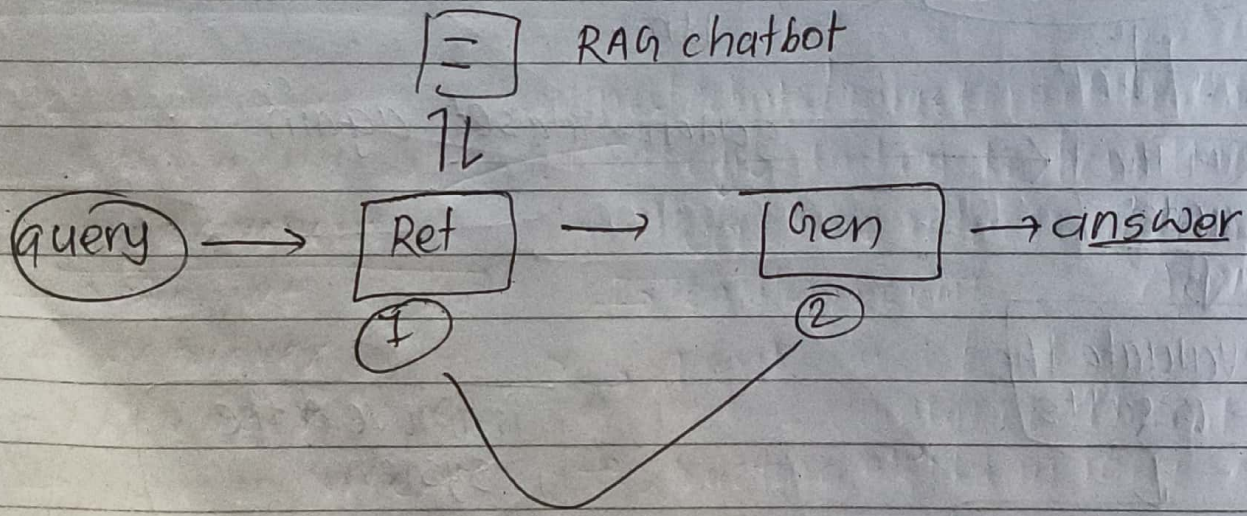
Deploy & Monitor → production failure

Improve the system



One LLM application have several LLM evals.

Why multiple eval Pipeline.



2 Failure Points.
So we need 2 LLM eval pipeline.

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Retriever eval Pipeline:

Given a query → are we getting right relevent docs.

Generator eval pipeline

Context → answer generated

faithfullness (groundness).

But does this guarenteed the RAG working fine??

Q9 what is the duration of ML Course?

K=5

Retriever:

d1: _____

d2: the duration of python is 4 week

d3: _____

d4: _____

d5: The duration of ML course is 8 weeks.

RAG eval ✓

System prompt:
higher doc - higher priority

context → LLM → the duration of ML is 4 week

Gen-eval ✓

Both Rag eval & gen-eval working fine.

Ret + Gen

A New eval is needed: Ret+Gen combinat eval.

Reranker

is there gurantee that IF Ret eval ✓
Gen eval ✓
Ret+Gen eval ✓
our app

work fine?

any problem

it this pipeline Takes 10sec to answer 1 question. Is this ready For production deployed?

Need one eval at application level

∴ Failure points/Target

Component
- prompt
- Retriever
- Reranker
- Query rewrite
- embedding model
- vector DB
- output parser
- tool selector.

memory
guard rails.

Workflow
- RAG workflow
- Agent ,,
- structure o/p.

entire Application

LLm Application eval - Risk categories:

Component → retriever → latency.

Workflow → cost effective

application → answer correct → safe//

3 Types of Risk categories:

Application quality → 'is the answer any good'?

safety → 'could the answer cause harm?

operational → 'is it fast, cheap & reliable enough to run'

different different eval pipelines:

Multiple eval pipelines Because

- Multiple failure points.
- Multiple Risk Categories.

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LLM Application Evals- Method

LLM output ⟶ Judge (Evals- Method)

1) programmatic / deterministic.
2) HUMAN
3) Model-graded (LLM-as-Judge).

---

eg Eval Pipeline → eval method → programmatic:

Chatbot → retriever

query

Building component wise eval pipeline. For retriever.

Define Task & Target

retriever

Component → retriever.

Define a success Criteria

→ out of all the correct / relevent items that exist, how many did the system retrieve in the top K results.

V-Imp ★

Recall@K = Number of relevent items retrieved in TOP K / Total no of relevent items.

= 1/2

Build a Dataset → golden Dataset ✓

↓

Define a eval method → Programmetically

recall @K

↓

Run the Model

↓

evaluate the results

↓

Analyze the result → improve the retriever

Eg Eval Pipeline → eval method → Human.

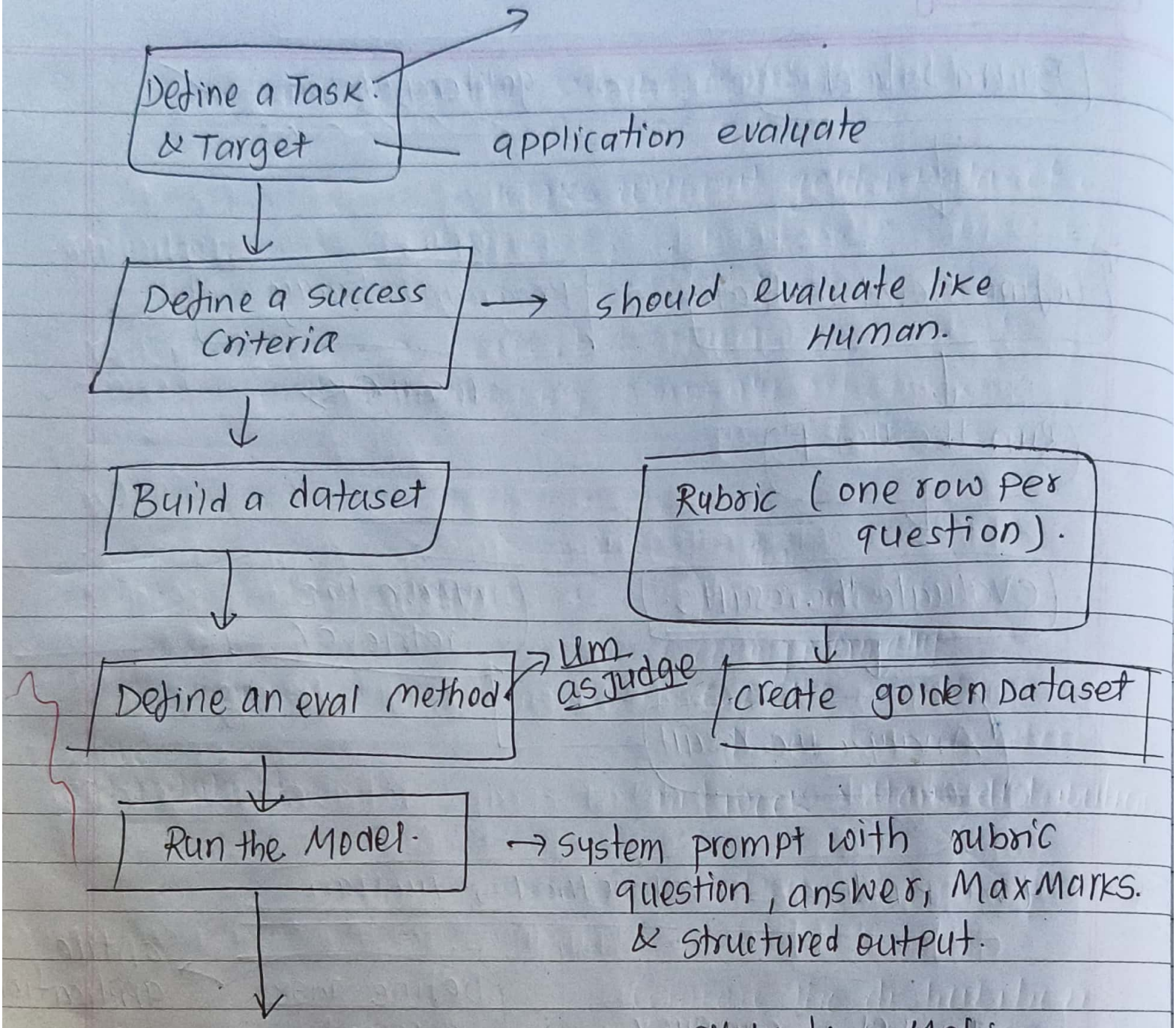
chat bot → campusx

define success - checking Helpfullness

Rubric: Helpfullness (1-5)

Define Task & Target → entire application

→ To evaluate helpfullness.



evaluation Metrics

Mean absolute error

Reference Based Evals	Reference Free Evals
We know answer in the golden dataset. → Llm as Judge + predetermine	~~only~~ when we didn't have answer on golden Dataset. → Human Based.

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V. Imp

Offline Eval V.S Online Eval.

Offline Evals vs online Evals. V.V.Imp.

↓

an eval pipeline put on the top of LLm Before Deployment.

↳ is it ready for the Deployment.

Why do we need offline Evals.

1. pre testing (Gating)
2. version Comparison.
3. Regression Testing.

But what about post-deployment.

1. Unanticipated inputs: – question mixing diff language, ambigious half question, injection attempt.

2. Emergent / systematic Failure ⇒ problem that exist only at scale.

(3) Drift

---

offline eval ⟶ obsolute

⑧ online Evals: → on production:

Works without golden Dataset.

to check.

offline Eval	→	correct working
online Eval	→	working Normally

Upsc grader

offline

↳ correctness.

same as the Human.

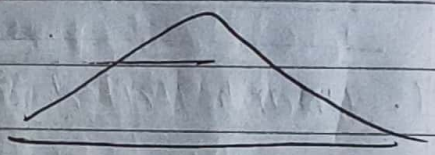
do have golden dataset

online

working Normally or Not.

↓

last week → Marks distribution



Compare the distrubition of score with previous week.

User Feedback

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online Evals pipeline:

step 1

Logging — V.V.IMP

→ Computed Signal
produced by an evaluator.

→ Langsmith

→ captured Signal
{Read directly from the Trace}.
eg: latency.

DashBoard → Alerting
/slack/email
/whatsapp.

captured quantity

log → dashboard → Alerting.

2nd

computed quantity — to check hallucination

logging → evaluator
Reference free evaluator

LLM as a Judge ← { question + retrieved context + output }

Rubric

But the problem is LLM as a Judge is costly.

Solution: sampling

logging → sampling → evaluator → Dashboard → Alerts.

But random sampling is not a good practice.

thumbs down + Rephrasing.

use stratified sampling

category with highly crucial & thumb down.

langsmith complete evaluation platform.

the self improving loop

offline evaluation

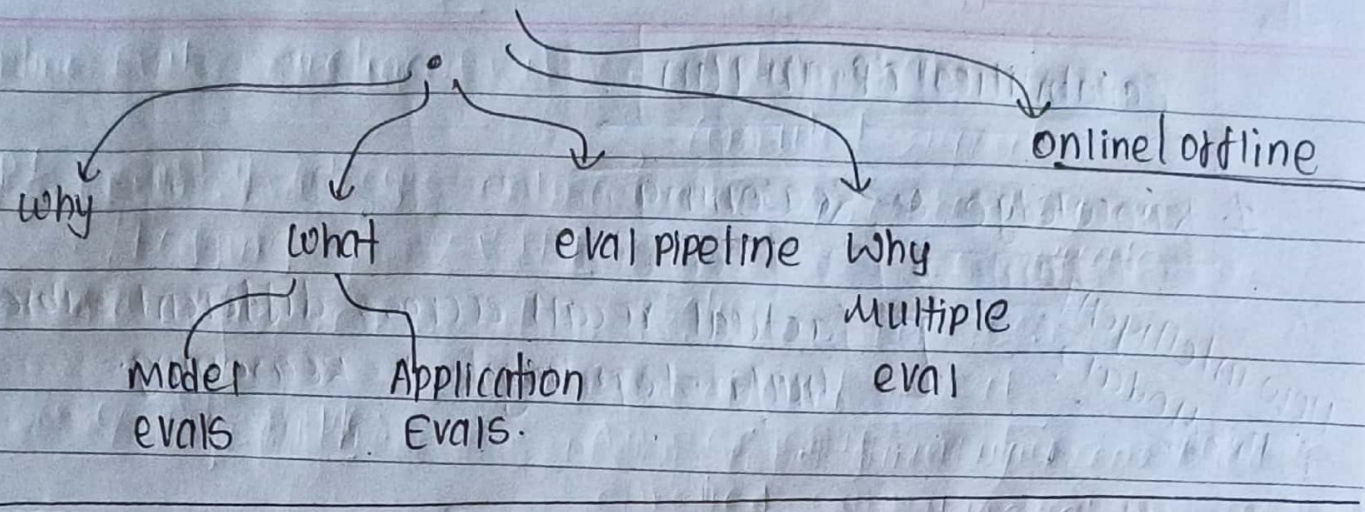
Production failure

↓

Add to the dataset

↓

~~Add~~ annotation queue



Model Evals

Why AI Engineer Needs Model evals → To Measure the capabilities

{ if you can't Measure you can't improve. }

1) To compare different models fairly.
2) To track whether newer models are Actually improving.
3) To decide whether a model is safe for release.
4) Build - versus - Buy decisions.

Model Eval

- Benchmarks
  - Math reasoning ✓
- Custom Evaluation.

Capabilities of an LLM

1. Knowledge & Reasoning

How intelligent is this Modal?

- factual recall across different subjects.
- multi step logical Reasoning.

2. Coding & Software Engineering — imp. economically

- Function level Code generation.
- existing Codebase Bug fix.
- API & functional calling.

(3) Mathematics:

Grade school mathematics.
Competation level.

V.Imp

(4) long context:

Single Fact Retrive From long context

(5) Vision & Multimodal

(6) Agentic & Tool use

(7) Saftey & Alignment.

9) instruction following.

Why Frointer labs evaluate this seperately

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Benchmarks:

↓

standerized test to measure a Particular model capability.

~~Daset~~ Dataset / Task.
Run Config.

GSM8K

Part 1 :- The dataset :

llm-evaluation harness.

eval harness. Built in library

Analogy

A Benchmark ⟶ exam Paper
and evaluation harness → exam administration system.

#

llm-eval-harness

eval-harness vs deepeval

↑ used for Application eval

Who runs Model evals?

1. Frointer labs (the Model builders).

Why they evaluate:

- ↳ internally, during development
- ↳ Release gating.
- ↳ Marketing.

How to read their numbers:

- ↳ lab numbers are always celling.
- ↳ Measured under favourable cond.
- ↳ Biased in selection & Configuration.

2. 3rd party evaluators.

independent organization.

LMArena.

leaderboard

Trust

they also measure cost & latency.

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3. Companies & AI engineering Teams.

Why BenchMarks Can be Misleading

1. BenchMark Contamination.

↳ public Benchmark.

↓

dataset → Publicly avaliable.

[Memorize] or [{ OpenAI - 5.6 ↳ recent all data on internet scrap } { Pretrained on that data }

2. BenchMark Saturation:

BenchMark 2021 — 25% 36%.

↳ static

Performance over these Benchmark work very good.

clustering at one place.

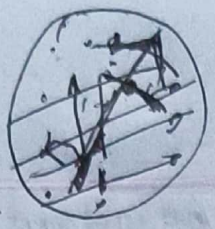
GSM8K

↳ Saturated Benchmark

(3). Configuration gaming

↓ favourable condition:

↳ run configuration. change



4. Aggregation:

MMLU → 57 subject question. → Physics ✓
→ economics ✗

They give you avg score.
Not individual performance.

#

Model evals

Benchmarks → Famous Benchmarks.
custom evals. → & capabilities.

10 Most popular Benchmarks.

Knowledge Capabilities → 7 Benchmarks.

evolution
↳ llm → train → retain

2020 onwards.

[Gpt-3] →

problem public

- MMLU (2020).
  - ↓
  - dataset (57 subject 14K mcq questions).
- Benchmark Contamination.
- Benchmark saturation.

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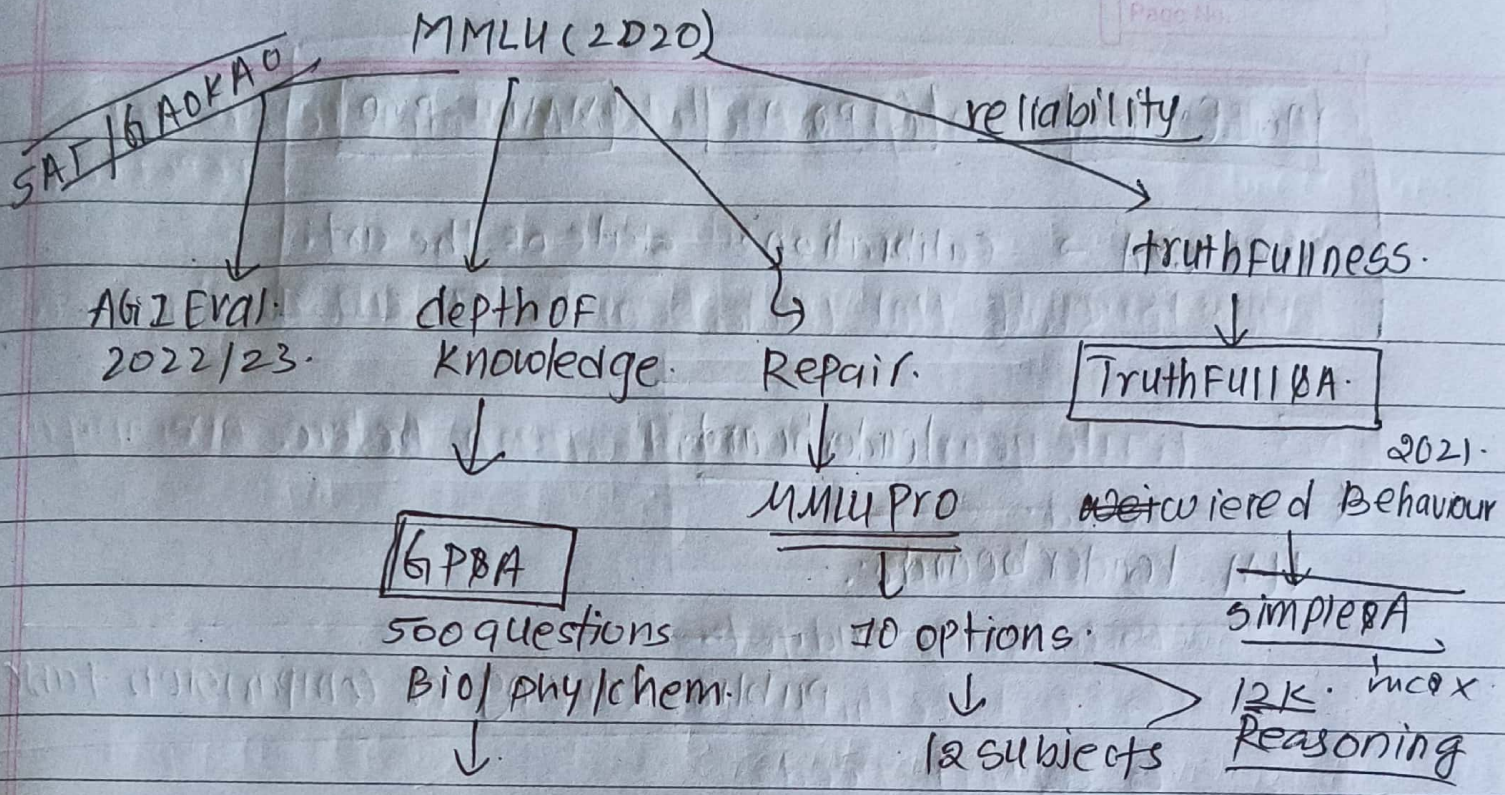

But with time they saturate:

Humanity last exam:

[HLE] ⟶ 2500 question.
100 subjects.

[depth + Breadth.]

Generation
MC1
MC2

llm as Judge

AGIEval · GPQA · MMLU-PRO

SimpleQA → Calibration · state of the art:

leaderboard

* LLM leaderboards.

Public Ranking or comparision table.

3rd Party

why leaderboard exist:

- ↳ provides trust
- ↳ you don't nave to run evals yourself
- ↳ Detect saturation.
- ↳ To discover new models.

who needs them

- AI engineers.
- Frontier labs.
- Researchers
- Policy makers & safety institutes.
- open-weight ecosystem. — discovery

Types of LLM leaderboards:

1. Bechmark-specific Leaderboard:
2. Multi BenchMark leaderBoard.
3. Human preference Based Analysis:
4. Application specific leaderboard.

V. Imp. you used the most.

Why can't you always trust LLM leaderboards.

↳ Benchmark performance may not transfer to real applications.
↳ Benchmarks contamination can inflate scores.
↳ Models can be over-optimized for the leaderboard.
↳ L.M arena

{ Good hart's law } principle ↗

When a Measure becomes a target, it becomes less usefull as a measure.

↳ overall rankings depends on subjective design
↳ small score difference may not be meaningful
↳ Rank Bias.
↳ Human preference leaderboards have human Biases.

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↳ The scores may be stale, incomplete or self reported.

How to Read LLM leaderboards as an AI engineer.

Step1: Write your requirements down before opening any board.

- ↳ Task Type.
- ↳ Latency budget.
- ↳ Cost ceiling
- ↳ Context needs.
- ↳ deployment constraints. on-prem?
  open-weights?
  data-retention?

Doing this First is what stops you rationalizing toward whatever is ranked #1.

Step2 pick the board that matches your task, not the famous one.

Agents → SWE-Bench Pro
Tool use → BFCL
chat → the matching LM Arena Sub-arena
RAG → MTEB
Budget-constrained → artificial analysis.

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Step 3

Read the leaderboard clearly:

↳ Don't Blindly Trust:

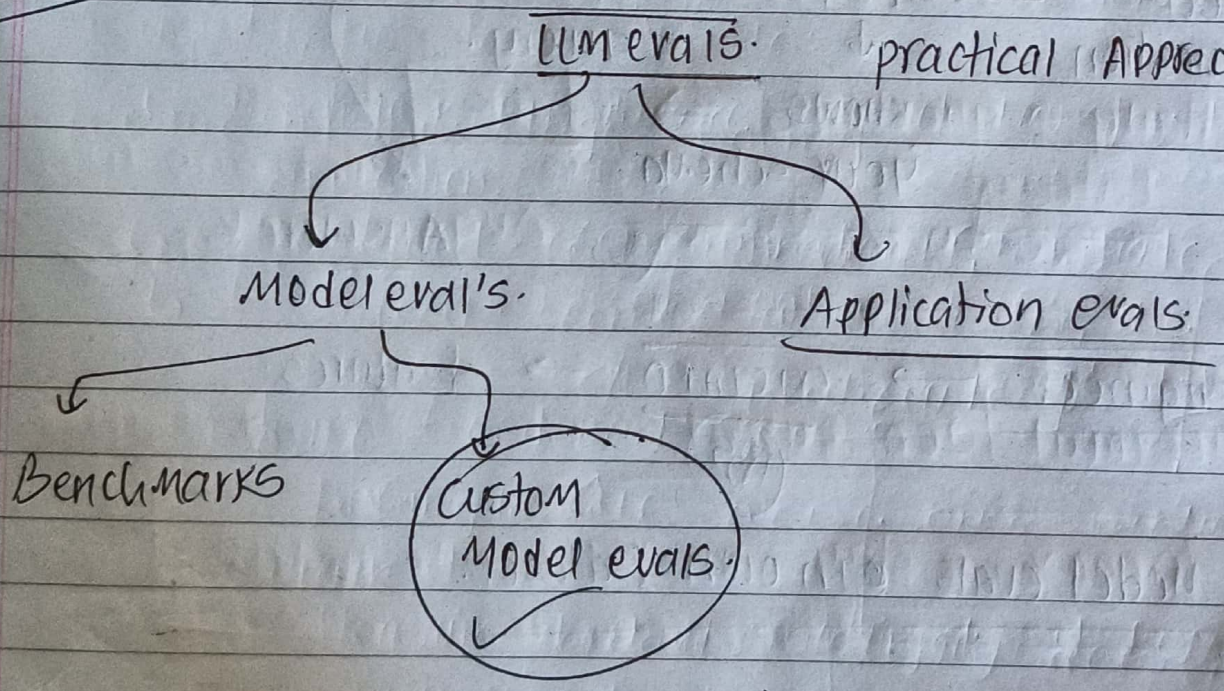
Step 4

Run your own evaluation.

* Leaderboards are a Filtering tool, not a decision tool.

Practical

LLM evals. practical Appreciate it.



Today goal

today class gist

How to run custom model evals:

Similar to

ESPN CricINFO

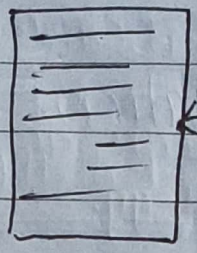
user query

analyst

DB query

Answer

TBS (Tokyo Broadcasting syste



scale.

AskCricINFO

text question → LLM → SQL query.

LLM ↓ knows your schema

Answer

AI engineer → CricINFO (TBS) → features.

Custom Model evals in own Dataset:

Text to SQL

Model

Flow

We use the LLM leaderboard to Filter

1) your requirements.
2) shortlist 5-10 models.
3) custom model evals.

Requirements:

1. task → Text2SQL
2. cost → Retention play → they want us to spend more time on website.
   ↓
   cost ceiling → 3 lakhs.

Text2 SQL system.

LLM → Per token cost.

Text → application → SQL query.
↓
Prompt

system:

You are a text-to-SQL generator. given a database schema return a single SQL query.

DB schema + query?

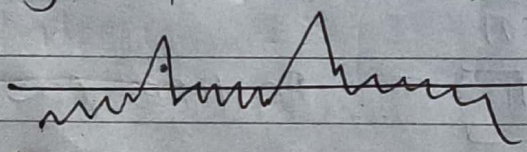
LLM charges for: input tokens + output tokens.

Calculate input tokens:
use token calculator.net.

≈ 352 tokens. ≅ 400 tokens.

Calculate output tokens:
use tokencalculator.net
≅ 100 tokens.

on avg daily → questions ⇒ 30K



Aug ≈ 50K

19 → 400 input | 100 output:

→ 50000 x 30 days x 95 USD to ~~INR~~ INR.

FableCost:

	input	output:
Claude Fables	10 $/M	50 $/M

Per query cost: $\frac{10}{1000000} \times 400 + \frac{50}{1000000} \times 100$

$\frac{4}{1000} + \frac{5}{1000} = 0.009$

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= 0.009 usd/query.

perday

=50K = 0.009 x 50K = 450 $ Daily

permonth

450$/Daily * 30 = ~~1800~~ 13500 $/Month.

13500 $/month * 95 (Conversion rate) = 1282500

But our Budget is only 3lakh/month. ≈ 4 times More Budget Needed.

Check Models of claude who have 1/4th the input & output token cost.

Refresh cycle = 5min or 1hr

Imp discussion

5m cache writes.	1h cache writes.	cache Hits & Refreshes
$12.50/MTOK	$20/MTOK	$1/MTOK.

prompt Caching

⇓

To save the input TOKEN.

Significantly Reduce in Cost:

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	Base input Tokens.	5M cache corites	1hr cache corite	Cache hits Refreshes
Claude Fables.	$10/M TOK	$12.5/MTOK	$20/MTOK	$1/MTO

you should know about all of above Terms.

How caching works:

KV cache

Attention mechanism

12 lakh to nearly 6 lakh.

3rd point

Latency ⇒ 2-3 sec. ⇒ if not OCD Triggers:

4th point: Context

↳ context coindow doesn't matter.

5th point Deployment

↳ you can use the API keys.

6th point Correctness

↳ Correctness Matters the most.

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step 1: 1st goal write Requirement :

gather requirement:

1. Task
2. Cost
3. latency
4. Context Window.
5. Deployment
6. correctness.

step 2 leaderboard (5-10) Models filter:

Check text to sql leaderboards → Not updated.
→ Don't Fully Trust ~~your~~ online leaderboards.

go to the coding leaderboard. as aliace.

llmstats.com they have good information.

Kimi K3

How to get the Blended number

$$\frac{10 \times 4 + 1 \times 50}{5} = \frac{90}{5} = \$1.8$$

1st step to choose models From LeaderBoard:

Cost ceiling 3L/month.

step1 Calculate Cost per Month. For models & eliminate models who will cost us more than 3L/month.

step2 Do Normalization of Rating for all the remaining models.

Val

So ~~min~~

So ~~MAX~~.

Normalize it [0 to 7].

step3 Normalize the speed:

$$\frac{Val - min}{max - min}$$

Normalize

step4

0.9 * normalized rating score

90% weight to coding capability

+ 0.1 * normalized latency score.

+ 10% weightage to latency score

Why 10% weightage to latency

~~to~~

SQL query hardly 10 to 20 lines.

So to write codeforces very easy for every model

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Now we will get our top 10 models:

	Model	Index	$/M0	throughput	Score
1.	GPT-5.6-Tera	46.0	4.47L	57 c/s	0.909
2.	Kimi K3	44.9	4.83L	4 c/s	0.883
3.	Muse Spark 1.1	38.3	1.65L	266 c/s	0.823
4)	Minimax M3	35.4	42K	171 c/s	0.761
	Qwen				
	GLM				

→ Open source Models

Do Custom evals on these models:

Test

1. GPT 5.6 Tera
2. Kimi K3
3. Muse Spark 1.1
4. Grok 4.5
5. Claude Sonnet 5
6. Minimax M3.

step 1: gather requirements.

step 2: shortlist 5-10 models from leaderBoard.

step 3: Custom evals:

① Dataset: → Kaggle IPl dataset

② we need golden dataset.

30-50 questions.

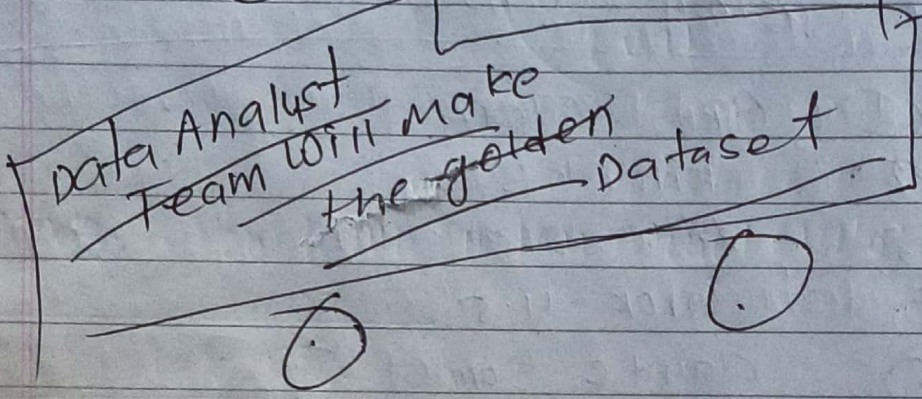
1 ——— , sql query

2 ———

⋮

50

10 easy questions
20 'Med "
30 hard "



③ evaluation with golden dataset.

q1 → 5-6 Tera → SQL → DB run. → llm query
→ golden DB query.

Result match.

~~write~~ Programmatically.
Check outside.

q2
⋮
950

True → 1
False → 0

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Now calculate accuracy For each model:

~~Now~~
How to prepare golden Dataset

Human Made → require money.
LLM calls → require money.

* custom eval:

✓ 1) The dataset → [ipl → 2020 – 2024.]

✓ 2) ↓ csv File → Fetch data & create DataBase#
~~2~~ → Filter & load the dataset.

✓ 3) extract the schema: [schema_extractor.py.]

④ Build the golden dataset → made with the Human experts.

Make 20 queries:

[golden_dataset_generator.py.]

GPT 5-6 Terq ——— 80
Kimi K3 ——— 55
Grok 4-5 ——→ 90
claude sonnet 5 ——— 85
Minimax m3. ←——— 65

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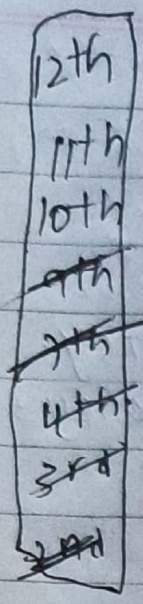


~~October 12~~ 1

LLMevals

Model evals

Application evals.

Benchmarks (Full theory)

Custom Model eval

Case study

Agenda: How to run Custom Model evals.

Problem statement:

Text2SQL APP → TBS (Tokyo Broadcasting System Television).

↓

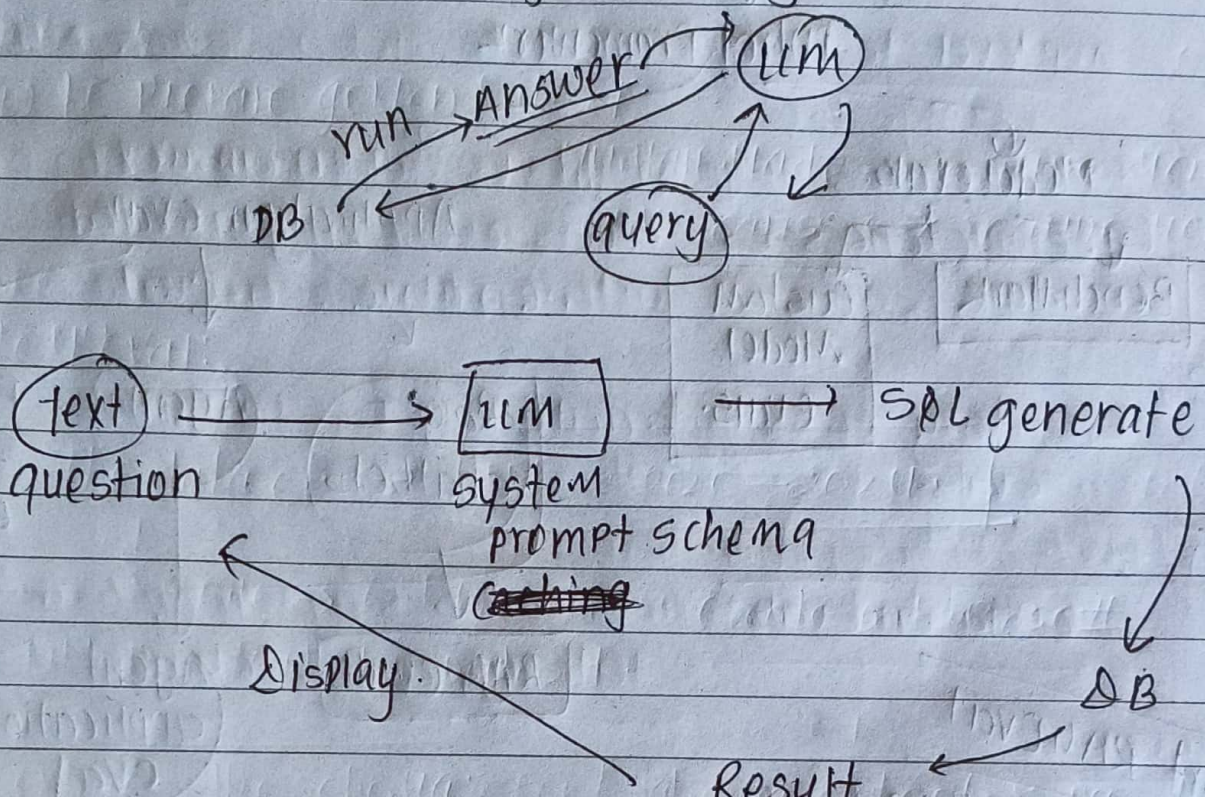
Domestic Broadcast right.

TBS → Platform streaming

↓

Add a feature Text2SQL at scale.

goal select suitable Best Model For Text2SQL with in Budget ceiling.



think??
Deeply

Why can't we use Best Models like Fable?

what about custom evals?

what about directly selecting from leaderboard?

LLM leaderboard Job is not to select Model.

it is used to Filter Models Based upon our requirement.

Select 5-10 models from leaderboard LLM stats & then run custom evals.

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